5.3 BIPOLAR JUNCTION TRANSISTOR

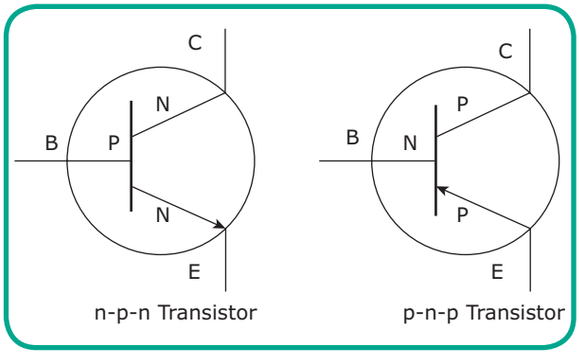


Figure 5.1 NPN and PNP Transistors

It is nothing but an ordinary transistor. These transistors are amongst the most widely used devices for amplification of all types of electrical signals and intermediate devices in discrete circuits, i.e., circuits made from individual components rather than integrated circuits. Transistors are also used in circuits together with integrated circuits, since it is often more practical to use discrete transistors where higher power output is needed, which is higher than the integrated circuit output.

There are two types of transistor

1. PNP Transistor
2. NPN Transistor

Remembering Tips:

P-Points	N-Never
N-iN	P-Points
P-Permanently	N-iN

Figures 5.1 and 5.2 show the symbol and structure of PNP and NPN transistors, respectively. In PNP transistors, a thin layer of N type is sandwiched between two P type layers. In NPN transistors, a thin layer of P type is sandwiched between two N type layers.

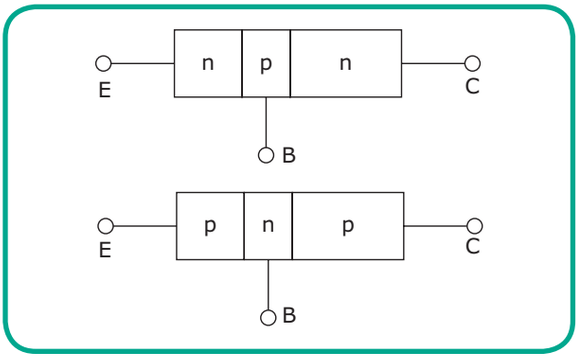


Figure 5.2 PNP and NPN Transistor

5.3.1 SMD Transistor

A SMD transistor is a type of transistor that is directly soldered on the copper points in the surface of printed circuit board.

Advantage of SMD Transistor

1. There is no need to make holes in printed circuit board.
2. Smaller in size.

5.4. SOME FACTS ABOUT THE TRANSISTOR

1. The transistor has three regions, namely emitter, base and collector.
2. The base is much thinner than the emitter while collector is wider than both.
3. The emitter is heavily doped, so that, it can inject large number of charge carriers (electrons or holes) into the base.
4. Usually the emitter is heavily doped, the base is lightly doped and very thin and the collector is moderately doped to collect majority carriers from the emitter.

ELEC_EQUIP_Chapter05.indd 100 11/10/2020 11:41:46 AM